

IQ Framework & Storage

Ephemeral

Usage

Durable

Cache Mgr/ Replacement Technique	Cache Mgr/ Replacement Technique
File System/ DBMS	File System/ DBMS

Volatile

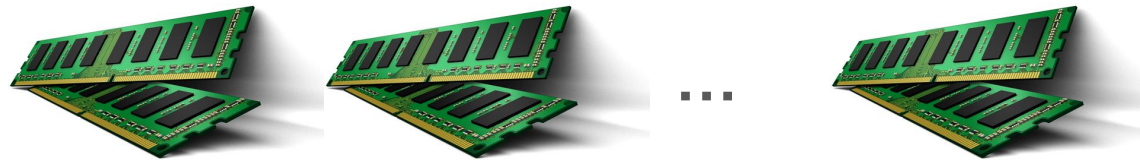
Non-Volatile

Type of Storage

RAMCloud

USC Viterbi
School of Engineering

DRAM
Storage Pool
(Durable)



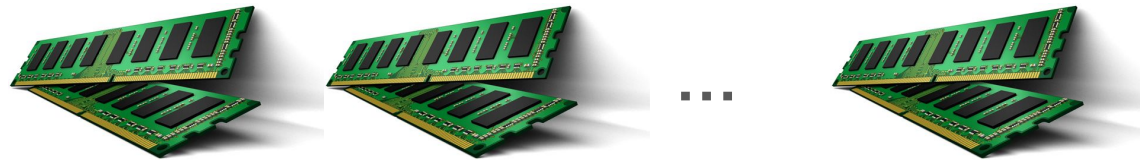
Key-Value Store



Ousterhout, John et. al. The RAMCloud Storage System. In the ACM Trans. Comput. Syst., September 2015.

Cache Augmented DBMS

DRAM
Storage Pool
(Ephemeral)



Query
Result
Cache

SSD
Storage Pool
(Durable)



Relational
Database

SSD
Storage Pool
(Ephemeral)

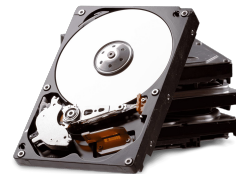
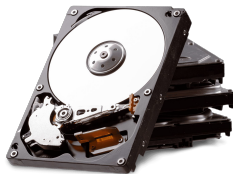


...

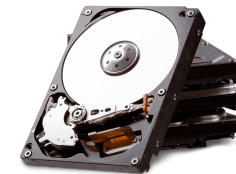


Query Result
Cache

HDD
Storage Pool
(Durable)



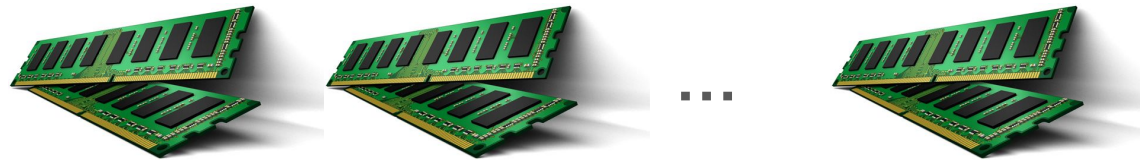
...



RDBMS (or
key-value
store)

Hierarchical Storage Structures Using Caches

DRAM
Storage Pool
(Ephemeral)



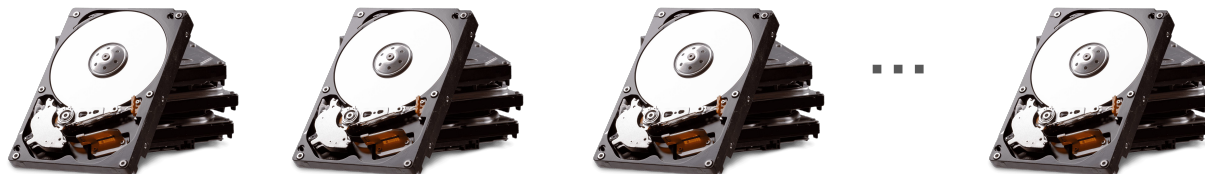
Query
Result
Cache

SSD
Storage Pool
(Ephemeral
& Durable)



Query Result
Cache
Relational
DB

HDD
Storage Pool
(Durable)



Relational
DB